Real-Time Sign Language Recognition and Translation System Using MediaPipe and Deep Learning

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Abstract

Sign language recognition has emerged as a highly significant assistive technology that has assisted in the termination of the communication barrier between the hearing-impaired people and the rest of the population. This paper will detail a Sign Language Recognition and Translation System, which is real time and built upon computer vision as well as deep learning to identify the hand gestures and translate them into translatable texts. The system presented implies the use of MediaPipe that identifies landmarks on the hand and finds 21 primary points of the hand and expects them to be recognized and processed by a trained deep learning model to recognize gestures. The described output is displayed through the functionality of a simple web-based application based on Flask, as well as turned into more than one language with the help of an in-built translation module. The system is capable of being utilized in real time performance with a normal web-cam and personal computer, although the performance of it can only be successful at a good condition of light. The low-cost method that is highly portable has been found to have the following practical applications in the learning institutions, service agencies and assistive communications devices and it can be used to bring the two communities (sign language and non-sign language) very much nearer. The availability of artificial intelligence combined with computer vision technologies serve to show how the assistive technology can contribute to the access and inclusion. Others are speech output, mobile deployment and full sentence recognition.

Keywords—Sign language recognition, computer vision, MediaPipe, deep learning, Flask, gesture recognition, assistive technology, real-time translation, neural networks, accessibility.

I. INTRODUCTION

The deaf and mute people use Sign language as the primary form of communication. Most of the individuals are however not proficient in sign language and it is a great challenge to communication and this affects learning, the ability to work, access to medical services and socialization. All these will necessitate the development of the assistive technology in the form of automatically recognizing the sign language gestures, and converting them into some form of text or speech.

A different alternative of Sign Language Recognition (SLR) is realized by computer-generated solutions, that is, identifying and understanding hand signs using cameras and machine learning models and is enabled by the current advances in artificial intelligence (AI) and computer vision. The systems can record the motions of the hands, retrieve significant features and classify it into a known category of signs. Deep learning and real-time video processing technologies have also increased the accuracy and performance of SLR systems and their accessibility.

The paper below shows a comprehensive SLR system utilizing the MediaPipe platform that is an efficient hand land mark detection solution and is utilized in determining 21 significant points along the hand with a high degree of accuracy. Here, the landmarks would be part of the input features of a deep learning model, which has been trained to recognize the different sign language gestures. The recognized gestures are displayed in the text form and presented with the Flask-based web interface assistance. It also has an inbuilt translation system which translates the identified writing to different languages such that the system becomes multi-faceted and useful.

The proposed solution will render it less reliant upon human interpreters and enable deaf-mute individuals to chat in real time with non-sign language individuals. It is also a cheap system which is portable and can be installed at various environments like at classroom, community service centers and smart assistive devices. This research can assist in unveiling how artificial intelligence can lead to viable applications of inclusiveness and access to supportive communication technologies.

II. RELATED WORK

Sign language recognition is a research area of interest. Computers, machine learning and human-computer interaction Computer vision Computer vision is a branch of computer science focused on perception, representation, and understanding of the surroundings via vision. <|human|>Human-computer interaction, machine learning and computer vision Sign language recognition Computers Computer vision is an area of computer science concerned with the subject of perception, representation and cognition of the environment through vision. The first systems were sensor-driven data gloves which were made up using flex sensors and accelerators to track actions of the hands [1]. These systems were correct but it was expensive, inconveniencing and could not be applied practically by the people.

The development of computer vision was as a result of gesture recognition systems made with cameras. The color of the skin, contours and the measures of shapes were the options used in such systems [2]. They were however quite sensitive to the factors of lighting, background noise and the quality of the camera used which compromised their reliability.

Machine learning and deep learning algorithms have been used to study gesture recognition in recent times. The image recognition of hand gestures by convolutional neural networks (CNN) is embraced massively [3][4]. The models are capable of automatically drawing spatial features, and are very accurate compared to other traditional image processing models. However, CNN-based solutions require too much data inputs and too much power to compute.

MediaPipe hand tracker has also been developed as a viable solution to applications in the real time mode [5]. It recognizes 21 hand landmarks free of latency and localization of key points. These landmarks can be utilized as feature vectors to be applied to the machine learning models during training hence eliminating the use of large-scale images. Other researchers have used MediaPipe in conjunction with neural networks, including ANN, CNNs and Long-Short-Term Memory (LSTM) networks [6][7].

However, along with these advances, existing systems have a variety of drawbacks, including the need to be trained at high training rates, incapacity to operate in low light scenarios, they must recognize dynamic gestures and be capable of multilingual translation, and are incapable of operating in the real-time using low-cost devices. The specified system addresses the following weak points: MediaPipe is applied to enable the work of hand landmarks recognition and deep learning model training on hand landmark features but no longer on actual images, which reduces the number of physical computations but multiplies the real-time speed.

III. PROPOSED SYSTEM ARCHITECTURE

A. System Overview

The proposed system architecture will have seven modules, which are linked with one another, but in their interrelation will provide real-time sign language recognition and translation. The system flow begins with the achievement of the video input, hand detection, feature localization, deep learning recognitions as the text output, language translation and presentations at the web interface.

The modular architecture also ensures scalability, maintainability and efficient processing. All modules possess functional specifications and it is being constantly combined with the rest (nearby) components thus offering the end to end processing capability of the capturing of the gesture and the translation of the multilingual text.

B. System Components

1 Video input module: It is a module that receives live video feeds through an ordinary web camera. Frame by frame processing is done to maximize the rate of video feed to a high-rate in order to support real-time performance as well as allowing sufficiently sufficient quality of images to identify hands.

2) Hand Detection Module: The given module will rely on the MediaPipe hand tracking solution and detect the hands in the image frame and generate 21 landmarks points that illustrate the palm positions and joints of each of the fingers. MediaPipe involves two pipeline stages which involve palm detection and hand landmark identification which ensures that the tool has high performance in various palm poses and orientations.

3) Feature Extraction Module: 21 coordinates (x, y and z) of landmarks identified by MediaPipe are transformed into a feature vector normalized. Normalization gives scale invariance and enhanced generalization of the model to the diverse size of hands and camera distance.

4) Deep Learning Classification Module: This corresponds to an implemented neural network architecture which receives the normalized feature vector as input and classifies in all the known sign language gestures. In the learning of the complex pattern of gestures, the model will employ the fully connected layers and appropriate activation functions.

5) Text Output Module: The gesture is the most likely label and it is converted to clear text and appears in the user interface. The list of gestures identified in this module is endless and one can always see the movement of own hand translated in real time.

6) translation Module: This will be a module which will translate the text discovered into other languages using the assistance of an API based translation service. The user has the option of selecting a language of his/her preference as a target language that enhances cross lingual communication and enlarging the scope of use of the system.

7) Web Interface Module: It is a web application built in Flask, which is expected to be easy to use and be referred to a standard web browser. The interface provides the user with a full-fledged experience with the live video feed, the detected hand landmarks, the recognized gestures text displayed, and the translated signal represented.

C. Hardware Requirements

It has designed the system to execute on off-the-shelf consumer grade hardware hence making it affordable and accessible. As follows are the hardware requirements: (i) Processor: Intel Core i3 or AMD Ryzen 3 or higher, (ii) RAM: 4GB minimum, (iii) Drive: 20GB free, (iv) Webcam: internal or external USB web camera capable of resolving 720p. The translation module will require the

client systems to have access to the internet and a web browser (Preferably Chrome) to operate.Fig. 1. Proposed system diagram of the real-time sign language recognition and video input display, MediaPipe hand detector, feature extractor, deep learning-based classification and the multilingual output user interface.

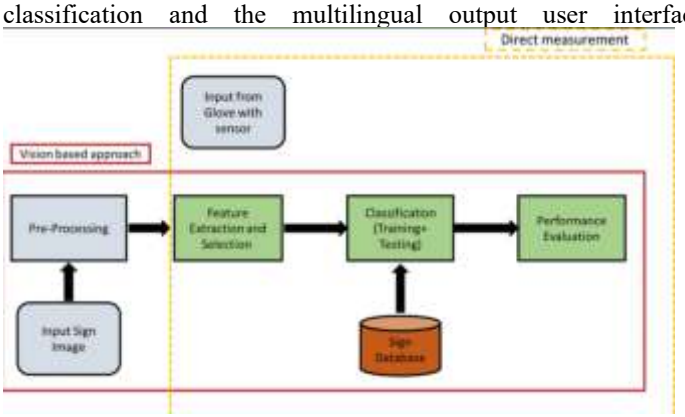

Fig. 1. *Proposed system architecture for real-time sign language recognition and translation showing video input, MediaPipe.*

D. Software Requirements

Its main programming language is Python 3.8 or later, and such indispensable libraries as OpenCV to process video, MediaPipe to track hands, TensorFlow/Keras to build and infer neural networks, NumPy to run number operations, Flask to create web applications, and Deep-Translator to assist in translation across multiple languages are used in the software stack. The universal environment facilitates the mainstream IDEs such as Visual Studio Code or PyCharm.

IV. METHODOLOGY AND IMPLEMENTATION

A. Data Collection

Data collection is one of the most important stages associated with gesture recognition that will be used to develop the appropriate gesture recognition model. The samples that are recorded with a web camera are characterized by hand gestures with every type of sign language. The gestures are repeated several times during different sessions because it is important to take into the account all natural differences in the position of hands and their orientation, as well as personal peculiarities of the user.

The procedure that will be used in the collection will be documenting the hand gestures in each category of signs, estimating the hand landmarks by the MediaPipe event in real-time, storing the coordinates of the landmarks as features vectors rather than images, and classifying the data into directories of the classes. This approach presents significantly less memory requirements and computational ability, compared to traditional image-based data sets.

B. Data Preprocessing

The pre-process of the data also improves the performance of the models since they do not alter the format of the input data and detect anomalies. The steps involved in processing a pipeline are as follows: making the landmark coordinates scale and translation-invariant through scaling by numbers and re-scaling by translations, eliminating noise, invalid samples caused by the absence of data, and converting the categorical labels into a format accepted by neural networks as well as splits the data into subsets (train and test) with the user-defined proportions (typically 80/20).

Normalization is done by computing the mean and standard deviation of the landmark coordinates in the training sample and standardizing them such that the means are zero and variance one. The step in preprocessing is useful in making the model learn useful gesture patterns despite the presence of any spatial differences.

C. Model Architecture and Training

The architecture designed using the deep learning model is an assembly of largely networked layers which will be trained to familiarize intricate non-linear associations between the locations of hand landmarks and the sign types of which they match. The input layer (that corresponds to the hidden layers with a gradually reducing number of neurons to enable hierarchical-based learning of features) is fed with the normalized feature 63-dimensional input vegetable (21 landmarks in 3 coordinates).

The multi-class classification of the training process is accomplished by the Adam optimizer that is employed to change the adaptive learning rate and the categorical cross-entropy as the loss function. Regularisation such as drop out also are implemented so as to achieve the elimination of overfitting. The model is trained in a number of epochs and has the early stopping functionality based on validation. Once the training is done, the model is then saved in the hierarchical data format (hdf5) in a.h5 file to be deployed.

D. Real-Time Gesture Recognition

Under the real time mode, the gesture recognition pipeline throughputs the system which captures video frames of the webcam continuously. The MediaPipe is capable of identifying hand landmarks on a per-frame basis and normalise them and afterwards, they are transmitted via the trained deep learning model so that they can be considered to be those.

The prediction process can be operated in interactive frames rates and it can be executed in 20-30 frames/second on the normal equipment. A small latency, that is reflected by the web interface, provides real-time feed-back to the users. This is attributed to the fact that such loop of processing brings about natural interaction between the users and system.

E. Flask Web Application

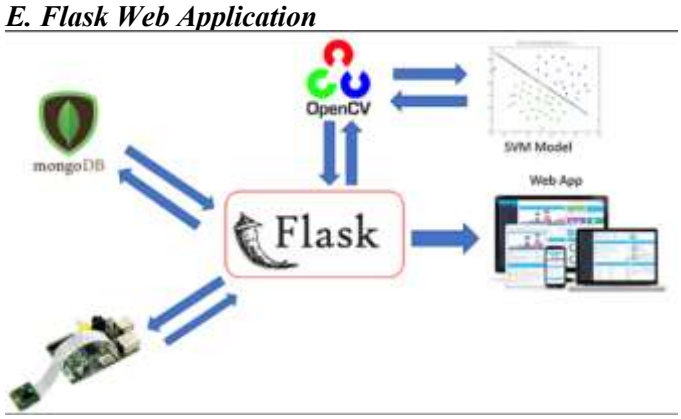

Fig. 2. *Web-based interface displaying live video feed,*

detected hand landmarks, recognized gesture text, and translated output.

The interface consists of a web application implemented on Flask which provides clear controls at the web interface such as the one that allows the user to activate the webCam and the real-time gestures, the text displayed, and the language options. The Flask server provides the opportunity to stream video through WebSocket connection, model inference and controlling the translation API. The friendly Web design can be equated to various devices and screen size.

V. RESULTS AND DISCUSSION

A. Performance Evaluation

The created system was prototested, regarding the accumulated sets of gestures, as well as the testing settings in real time. The trained deep learning model achieved great accuracy when it came to classification applied to test set proving that it was learning or gesture pattern. Live testing has shown that the system picks the gestures in a fraction of a second which is the need of an interactive application.

The system is able to be effectively utilized when the conditions in an indoor environment are normal with minimum complexity of the background. The computational cost of the landmark-based feature scheme on top of raw images is significantly less and its use can be strain-gated to real-time on off-the-shelf consumer hardware with no requirement of dedicated GPUs.

B. Translation Module Evaluation

Its inbuilt translation service has been capable of translation of the known English texts to other target languages like Spanish, French, German, Hindi among the rest. The quality of the translation work is dependent on the supporting API service and it is in line with the conventional machine translation (system) in the industry. This module is also marked by the low latency which is usually completed in less than 100- 200milliseconds of translation.

C. User Interface Evaluation

The Flask Web GUI has easy to use interface with properly organized layout segment of live video feed, hand landmark visualization, and know gestures text and translated output. It is touted to be a user-friendly interface and the user feedback indicates that it requires very low technical skills to use.

D. Limitations

Though it has proven to be a valid system in the event of stationary hand gestures, several limitations that were introduced. The poor performance occurs due to the low light or when the background is complex such that the hands are hidden over it to be identified. As it is currently implemented, the current system is confined to the stagnant gestures and requires enhancement to implement moving and flowing sign language orders. Besides this, the gesture language is limited to the trained categories and it might be improved to incorporate the holistic sign language alphabets and phrases.

Practical Applications and Societal Impact

Connection with the practical significance of the suggested sign language recognition and translator system that will work in the real-time has a high level in the spheres of real-life activity. One of the best applications is in schools and colleges where hearing impaired students tend to be at a disadvantage in communication between the students and the teachers hence not allowing them to sleep well at school. By using gesture-to-text translation, real time translation method, the system will assist the teachers to comprehend questions and answers of the students, even though the professional interpreter is not present. This results into inclusive classes as well as providing equality in classrooms.

In a healthcare facility, effective communication is a prerequisite in proper diagnosis of the patient and in giving care to the patient. The deaf and mute patients tend to have difficulties in defining of the symptoms and interpreting of the medical orders. The proposed system must have the capability of being an assistive communication system in hospitals, clinics, or telemedicine systems. When the sign language is translated in a readable form providing medical personnel with the opportunity to know the needs of patients in real time, it becomes much easier to increase the quality of provided care and reduce the number of risks related to the lack of understanding.

The system can also be applied much in the services sector of the government such as banks, government offices, transport centres and customer support centres. In such environments, limited or little communication may exist between the service delivery agents and the hearing citizens that will be based on a written note which may not be that effective. The feature that enables real-time translation of the system makes it easier and ensures that all important services of the system can be used by any person and users will not encounter difficulties accessing or using them, regardless of the power of their communication.

The system can also assist the employment inclusiveness as far as work places and professionalism are concerned because the deaf and mute is likely to interact more with the other workers, supervisors and clients through the system. This system can assist the organization to serve the purpose of closing communication gap in the organization, enhancing work team; and increasing workplace diversity through inclusive hiring policy. Additionally, it is also applicable to the virtual meeting by being connected with video conferencing software, which will facilitate the translation of the sign language remotely and, consequently, its subsequent extension to the modern hybrid work environment.

The other significant application is within the social communication and in our day-to-day interaction. The hearing-impaired individuals feel confident in participating in the activities of the society through the system in regards to shopping, traveling and other social activities. It promotes independence and blending at the social level by doing away with the necessity to talk with an interpreter or written form of communication. Furthermore, it is also supported with the multilingual translation which would enable interaction across the linguistic borders which would make it particularly useful in multicultural groupings.

In terms of technology, the system reveals that it can be possible to combine computer vision, deep learning, and web technologies in one assistive interface. The MediaPipe implementation entails hand landmark detectors that are significantly less computationally

intensive than image-based detectors, and can run on a cheap consumer-friendly hardware. This affordability is quite fundamental to apply in the creation of components, where availability of better assistive technology is generally limited.

Even though such systems have been proved to be positive, they have caused serious ethical and social problems in instances where they are widely applied in the society. This should take the privacy of information and consent of the user into account in real-time video feed recording. The system shall not violate the law of data protection as well as hold any sensitive visual information without their express consent. Furthermore, the developers are to ensure that the training samples consist of a diversity of hand shapes, and the diversity of the skin tones and styles of signature in order to eliminate the bias in the algorithms and in order not to give the users of different groups unfair performance.

The system will also assist in the internationally established goal of universal access and this will accompany the principles in the United Nations Convention on the rights of persons with disabilities (UNCRPD). The proposed model will reduce the digital divide and contribute to an equal role in the society by providing the technological tool that enhances the availability of communication.

Concisely, the resulting system is a technological breakthrough, not to mention, a socially important assistive system. Its applications are in the education, health care, state and local government and employment and everyday communication. As it further develops and gains more and more implementation, it can significantly improve the life of people with hearing impairments as well as increase the accessibility and inclusivity of the society.

Robustness Optimization and System Evaluation Framework

To make even the proposed sign language recognition system more credible and practical they introduce more optimization schemes and evaluation systems. These advances will work toward realization of elevated extents of robustness to perform in varied environmental and generalization of the designs and offer common performance appraisal organization.

The aspect of environment variability light condition, camera angle and the complexity of the background is one of the significant barriers of computer vision-based gesture recognition systems. The system has dynamically variable preprocessors to address this like dynamic background suppression filters and dynamically changing brightness normalization filters. The tools of processing maximize the intensity between the region and the surrounding areas, which enhance the more landmarks are detected. It is also through the use of data augmentation techniques such as rotation, scaling and small perturbation of coordinate on the training dataset to generate and improve the generalization of the model discriminative of a large variety of real-world scenarios.

The other worthwhile addition is the element of homogenizing the time and stabilizing the forecasts. Customized priorities of landmark detection in the real-time streams may cause discontinuity in the prediction of adjacent frames. To mitigate this issue, a prediction smoothing of sliding window is embraced. The approach not only stores prediction results on a short sequence of frames but also adopts majority voting or weighted averaging to make a final decision on the output. The method assists in raising the noise in the classification and reducing stability and reliability of gesture recognition.

The deep learning model of the system also contains aconfidence scoring mechanism. The output layer score of the softmax of each of the predicted gestures is confidence probability by each of them. Only the predictions that exceed some pre-established confidence levels are read by the user and the uncertain predictions are ignored or highlighted. This system removes the risks of impropertranslations and it increases the user confidence of the system.

To provide the systematic evaluation of the system performance, a full assessment framework is created. The model analysis is conducted in terms of the classic classification metrics (e.g. accuracy, precision, recall and F1-score of each of the gesture categories). The confusion matrix is developed to investigate the performance within classes and to know the gestures that are most commonly misclassified. In addition to offline testing, real-time testing is also performed to find out latencies, frame rate (frames/s) and end-to-end response time using gesture input to translated output.

Another way of usability assessment is usability testing, which is user-centered. Figure 1 indicates that participants, who may also be a sign language user and non-user, communicate with the system in controlled situations. The responses will be obtained in terms of ease of use, response speed, and attitude towards correctness and clarity of the interface. The interface design is changed according to this qualitative feedback to fit with the interface design and the overall user experience.

At the deployment end, the system can be compatible with the edge computing environment through optimization. The system can optimally run upon the constrained computing devices, e.g. laptops and embedded systems, due to the architectures of lightweight models and optimized inference pipelines. It makes the solution scalable and can be implemented on remote or resource constrained environment.

Besides, the system is expandable, depending on designs with modules, and more sophisticated components such as sequence-based models (LSTM/Transformer architecture) may be incorporated, such as speech synthesis capabilities, and offline piping of translations. The modular architecture simply establishes the fact that the system architecture can be extended without necessarily engaging in the full redesign.

Overall, these undergirding of strength and assessment strategies are long in bringing out better functionality, dependability, and ease of using the system. Based on practical case studies and having formal strategies of validation, the proposed solution may turn into something more than a prototype and the deployable assistive communication platform, which will be capable of helping to provide the inclusive and accessible human-computer interaction.

VI. CONCLUSION

The present paper presented a Sign Language Recognition and Translation System in real time, with the assistance of MediaPipe hand landmark detector and gesture decoder on deep learning. With the assistance of automatic gesture perception and multilingual

translation, the system can close distances present between mute individuals with deaf and the rest of the population.

The advantages associated with the proposed solution include cheapness through the use of generic equipment, cross-platform portability, live functionality which might be used in case of interactive applications and multilingual compatibility which can make it easier to use. The system reduces the computational intensity and has high accuracy through landmark based feature extraction.

The integration of computer vision, deep learning and web technologies is the example of how well the assistive communication industry can use the artificial intelligence. This research will help in the accession and accessibility of the hearing impaired within the learning units, the place of work and within the society.

VII. FUTURE WORK

There are several enhancements in the research direction in future which can be made to the current system. Introduction of sequence-based deep neural networks such as the Long Short-Term Memory (LSTM) or Recurrent Neural Networks (RNN) would enable seeing dynamic gestures and the sentence flow on a sign language. A text to speech program would be included to aid in communication with the sightless as well as enable the production of the audio.

Mobiles Lightweight deep learning models would result in more portability and the ability to access smartphone/tablet when deployed on mobiles. It should also broaden the gesture vocabulary quality into full-scale sign language dictionaries and parallelism to several sign language standards (ASL, BSL, ISL) to make it more feasible. The methods of superego image enhancement would be more effective in bright conditions and complexes in the background.

Other additions include multi-hand gesture recognition, real-time sentence structure using the Natural Language Processing, deploying to the cloud in order to make the product more scalable, offline translation and support to Internet of Things device including smart assistive environment. Such developments would make the system a multimodal and multi-lingual assistant communication system.

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